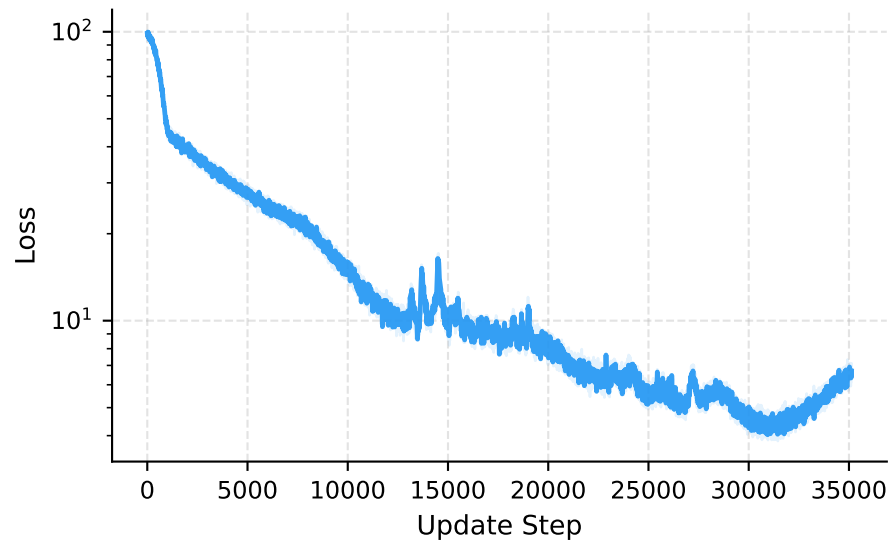
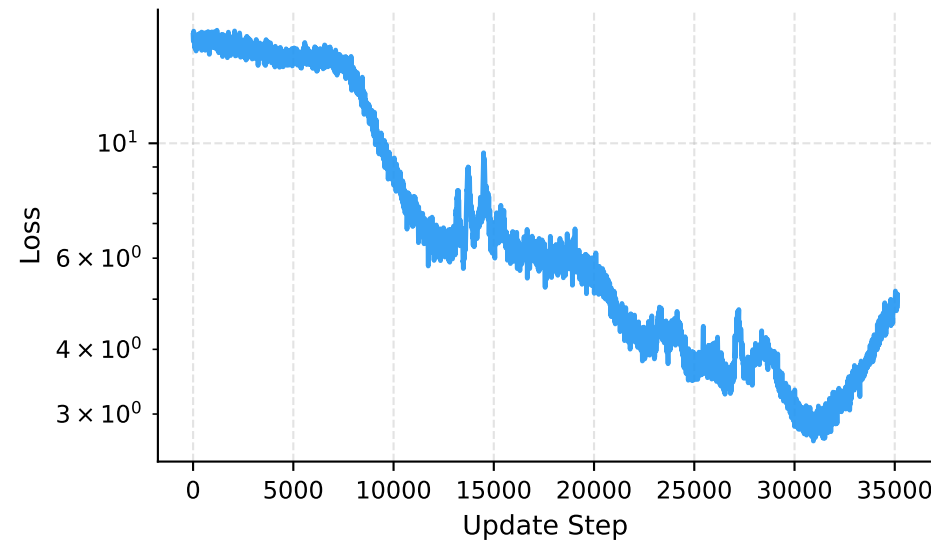


SpeciesLLM Training Curves
min_lr=1e-07 warmup=0.1 bs=512 gas=1 ep=5 seq=640

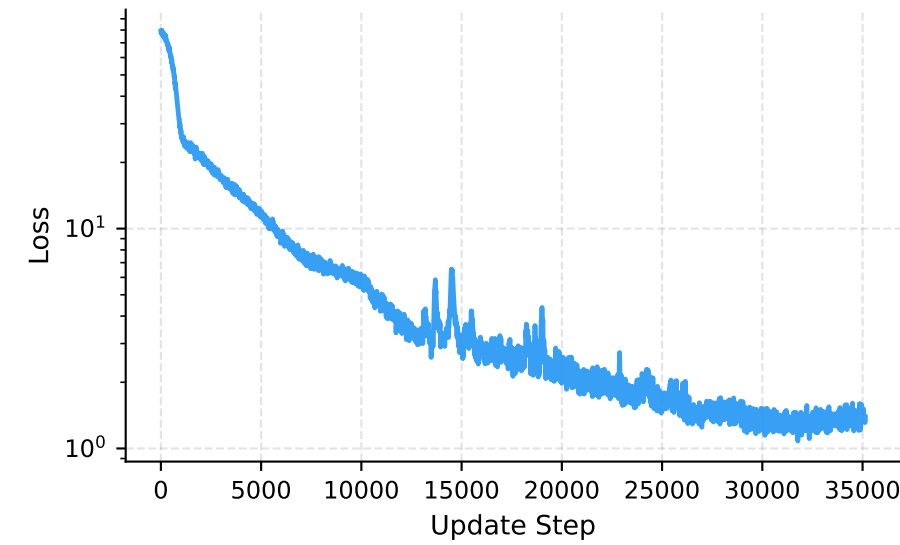
Total Loss



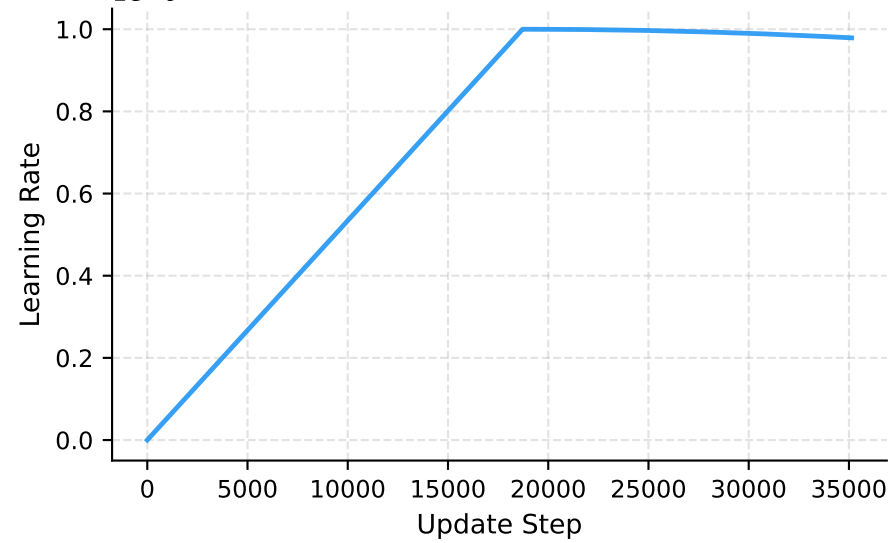
GEP Loss



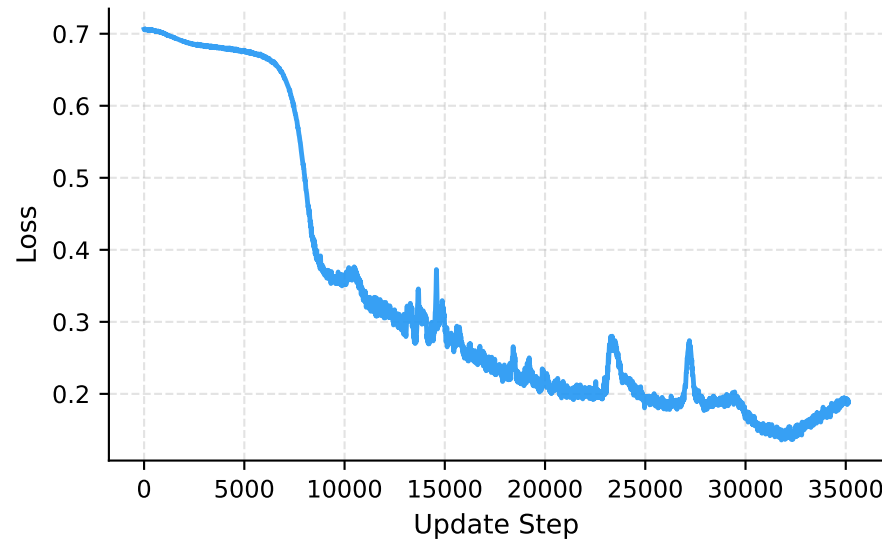
GEPC Loss



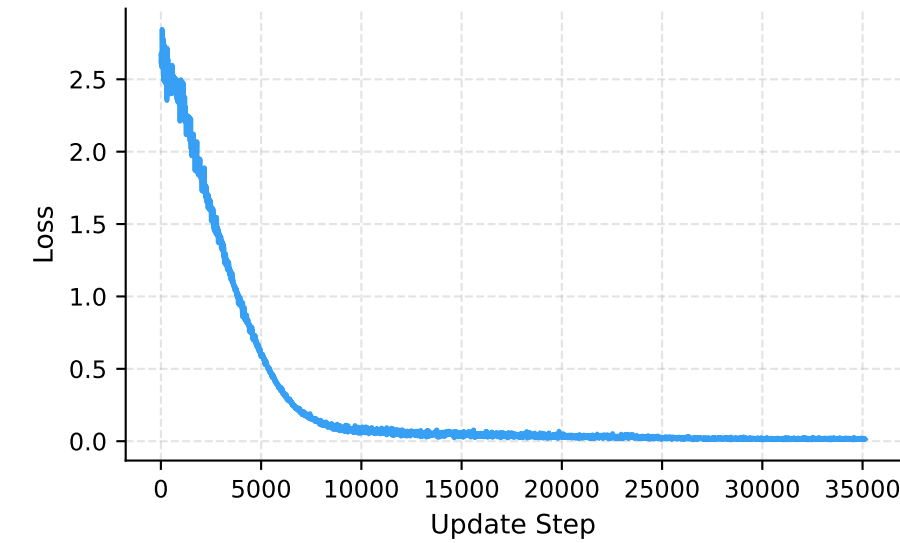
LR Schedule



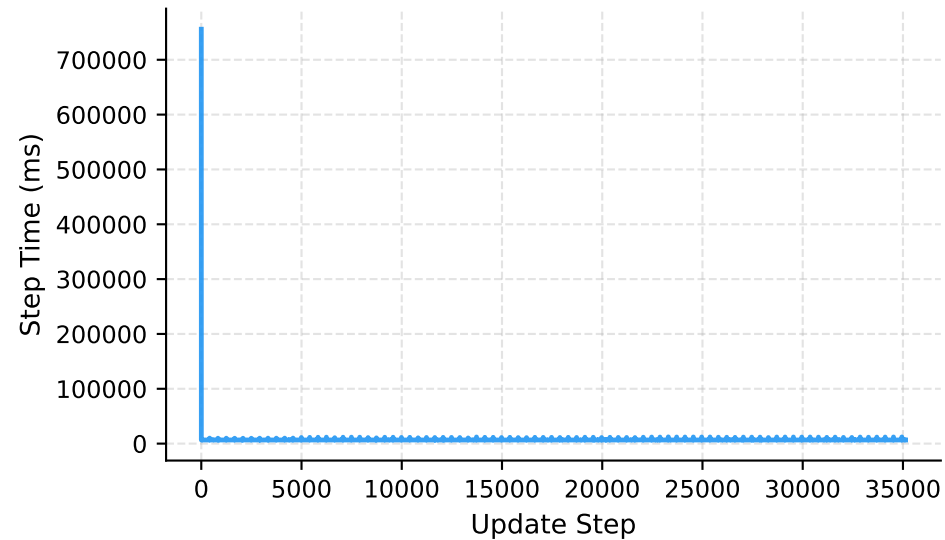
Zero-Prob Loss (GEP)



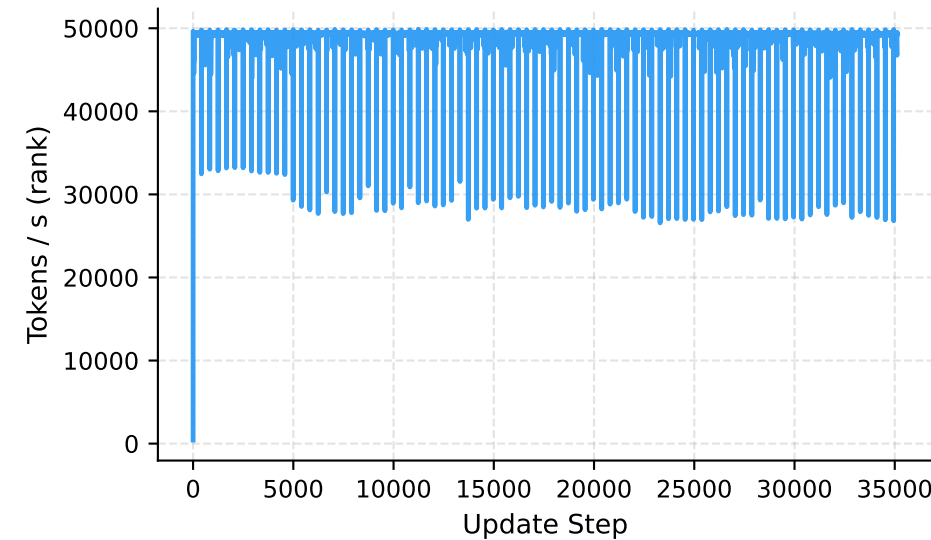
Zero-Prob Loss (GEPC)



Step Duration



Per-Rank Throughput



Cluster Throughput

